EE6083 VLSI Testing

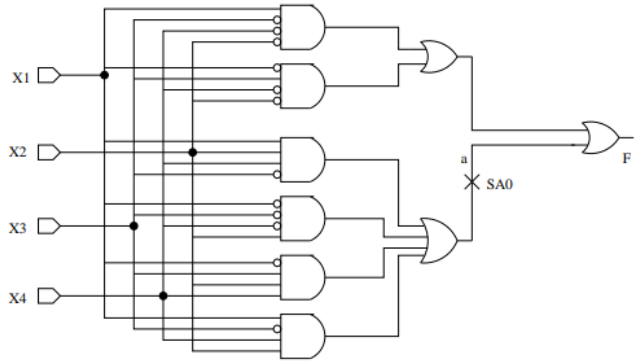
HW2

1. **Signature Analysis**
   Suppose that a modular LFSR with a characteristic polynomial $1+X+X^4$ is used as a signature analyzer for the circuit under test shown below. Please complete the following questions.

   (a) Use Verilog code to realize the LFSR and simulate the signature of the good circuit. (4 points)

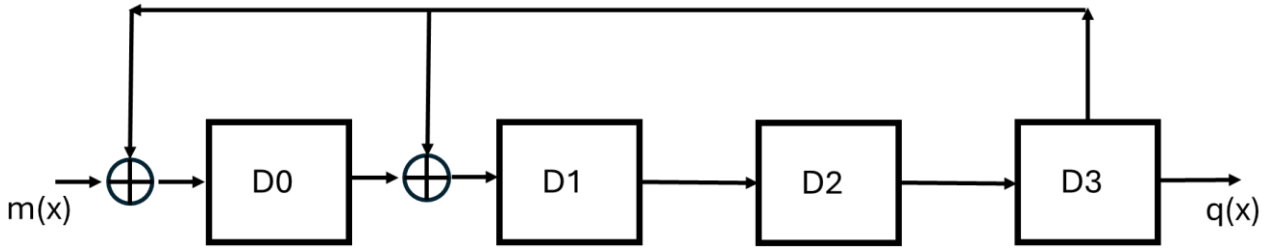
   (b) Let line a have an SA0. Simulate the signature if the test sequence is 1,2,3,...,15. (2 points)

No.	X1	X2	X3	X4	F
1	0	0	0	1	1
2	1	1	0	0	1
3	0	1	1	0	0
4	0	0	1	1	0
5	1	1	0	1	0
6	1	0	1	0	0
7	0	1	0	1	0
8	1	1	1	0	1
9	0	1	1	1	1
10	1	1	1	1	0
11	1	0	1	1	1
12	1	0	0	1	0
13	1	0	0	0	1
14	0	1	0	0	1
15	0	0	1	0	0



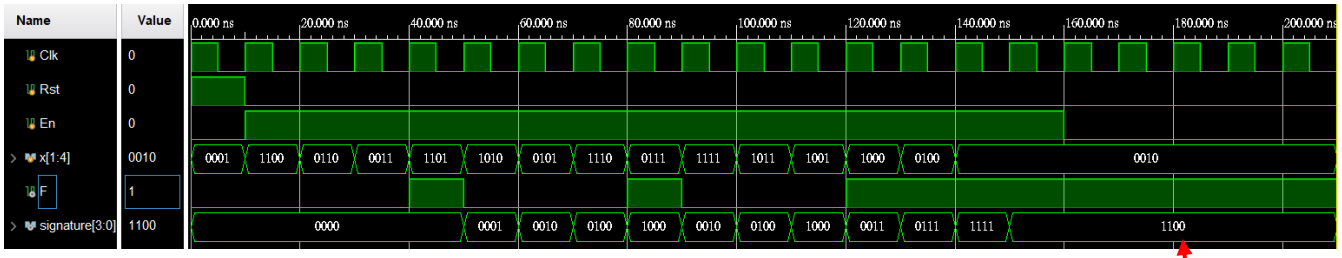
Sol.

第一題為設計 LFSR 電路，根據其 characteristic polynomial:$1+x+x^4$，LFSR 電路如圖一所示:



圖一、LFSR 電路架構圖

(a) Golden signature:



圖二、(a)小題 golden signature waveform

根據題目一的 15 個 pattern 和待測電路，經過 LFSR 後得到 1100 的結果相

Golden signature

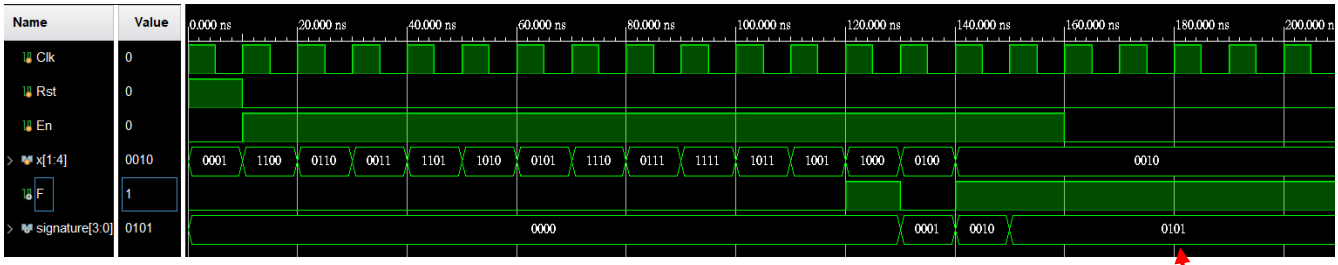
當於x^3+x^2與直接透過手算得到的結果$r(x)=x^3+x^2$一致。

$$m(x)=c(x)\times q(x)+r(x)$$

➔

$$F\equiv m(x)=x^{10}+x^6+x^2+x+1=(x^4+x+1)\times(x^6-x^3+1)+(x^3+x^2)$$

(b) SA0 signature:



圖三、(b)小題 SA0 signature waveform

假設待測電路有 SA0 fault，經過 LFSR 後得到 0101 的結果相當於x^2+1與直接透過手算得到的結果$r(x)=x^2+1$一致。

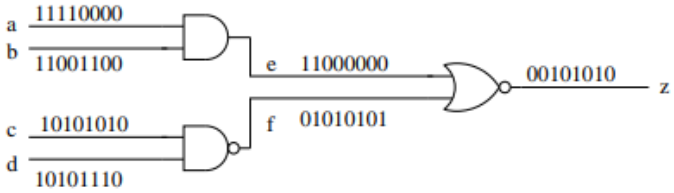
$$m(x)=c(x)\times q(x)+r(x)$$

➔

$$F\equiv m(x)=x^2+1=(x^4+x+1)\times(0)+(x^2+1)$$

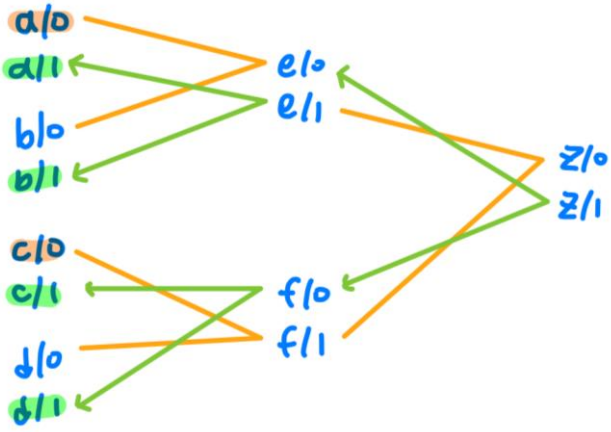
2. **Syndrome Compaction**

For the circuit shown below, use the stuck-at test given and find the signature of the good circuit if the time compaction used (a) Parity testing; and (2) one's count. For each scheme, list the undetected faults and explain why they were not detected. (4 points)



Sol.

先根據 Fault Diagrams 進行 Collapsing(圖四)得到 fault list = {a/0, a/1, b/1, c/0, c/1, d/1}。



圖四、Fault Diagrams

表一、Signature of Fault list

	No fault	a/0	a/1	b/1	c/0	c/1	d/1
z	0010_1010	1010_1010	0010_0010	0000_1010	0000_0000	0010_1110	0010_1010
Parity testing	Odd	Even	Even	Even	Even	Even	Odd
One's count	3	4	2	2	0	4	3

從表一的結果顯示只有 d/1 fault 無法被 parity testing and One's count 兩種 syndrome compaction 偵測到，推測其原因為缺少(c, d) = (1, 0)這組 pattern，所以 d/1 的 z 其實和正確結果一樣。

感謝助教仔細批改!!!